

The AI-coding landscape

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That is powerful, but of course also dangerous.

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One way to measure this is **task completion time horizon**: how long a task can be while the model still succeeds at a given rate.

Recent estimates suggest this horizon has been increasing roughly exponentially.

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The lesson

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What matters more:

- what kinds of help these tools can provide
- what kinds of mistakes they tend to make
- how to structure your work so mistakes are caught
- how to keep yourself in control of the scientific reasoning

Worked example: one Gemini call

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This was sent to Gemini using the same REST pattern from the API chapter, with `GEMINI_API_KEY` read from the environment.

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```
import math

def fisher_z_transform(r):
    """
    Calculates Fisher's z-transform of the correlation coefficient 'r'.
    """
    if not (-1 < r < 1):
        raise ValueError("Correlation coefficient must be between -1 and 1.")

    return math.atanh(r)
```

Gemini response, abbreviated

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This is basically correct, but we are not done!

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- `r = 1` and `r = -1` raise clear errors
- non-numeric inputs fail predictably

Try the same prompt again:

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```
body = {  
  "contents": [{  
    "parts": [{  
      "text": "write a Python function that returns Fisher's z transform of r"  
    }]  
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Try the same prompt again:

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You may get a different implementation, that's expected!

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The last point matters directly for coding! Wrong code often looks like right code.

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LLM hallucinations always looks good in context. As behavioral scientists we should probably call them “confabulations” instead.

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For scientific code, this seems like an excellent bargain between AI and human:

- AI can accelerate draft implementation.
- Tests, review, and domain knowledge decide whether the draft survives.

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Like financial debt, technical debt is not automatically bad – sometimes you borrow on purpose, to ship a result before a deadline or to keep a script readable while you're still exploring.

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The bottleneck in scientific programming is not the typing but the understanding!

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- asking for a function draft
- using autocomplete in an editor

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The model is not just writing text, but is acting in a development environment and iterating on its own output.

Worked example: an agent loop

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```
while not tests_pass():  
    proposed_patch = model.generate_patch(  
        instructions=task,  
        context=current_files(),  
        last_error=test_output(),  
    )  
    apply_patch(proposed_patch)  
    run("python -m unittest discover -s tests")
```

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More autonomy → stronger guardrails.

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- **Test constantly**
 - the agent can still make a mess of your repo

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- the tool exposes a standard interface
- the agent can ask to use the tool when needed

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Example MCP tools

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MCPs add **productive constraint**.

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You make a **schema** to describe the tool's arguments and return value. The host can validate arguments **before** running anything.

Example: Tool definition

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```
{ "name": "d_prime",  
  "description": "SDT d' from outcome counts.",  
  "inputSchema": {  
    "type": "object",  
    "properties": {  
      "hits": {"type": "integer", "description": "N hits"},  
      "false_alarms": {"type": "integer", "description": "N false alarms"},  
      "n_signal": {"type": "integer", "description": "N signals"},  
      "n_noise": {"type": "integer", "description": "N non-signals"}  
    },  
    "required": ["hits", "false_alarms", "n_signal", "n_noise"] } }
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The user asks: “Given 7 hits out of 10 signals and 2 false alarms out of 10 non-signals, what is d' ?”

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Same tool can be used by any MCP client that speaks the protocol.

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The model reads that result and writes the final answer to the user.

Servers answer tool calls

The server computes d' from $H = 7/10$ and $F = 2/10$ and returns a structured result:

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  "content": [  
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Official overview: modelcontextprotocol.io.

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(We'll soon talk about how to wire MCPs up for Cursor.)

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After `pip install -r requirements.txt` in a venv, `python client_demo.py` should print `tools: ['d_prime']` and the `d'` value for 7/10 hits and 2/10 false alarms.

Four interaction patterns

Pattern	Typical use
Single-turn generation	Draft one function, regular expression, SQL query, or explanation
Context-aware completion	Fill in code inside an editor using nearby files as context
Programmatic access	Call a model API from Python, as in the Gemini example
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These are not strict categories, most current tools blur them.

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Pattern 2: context-aware completion

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It is risky when the local pattern is wrong – AI is **great** at imitating bad examples. It's also **wildly annoying** when it keeps trying to autocomplete based on some misunderstanding.

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A model API is still just an API!

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- remembering every syntax detail
- writing boilerplate by hand
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More important:

- decomposing the problem
- defining success
- reviewing code
- testing behavior
- recognizing when the model is going in circles